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How to Cipher Like a Soviet

See if you can figure out how the American code-breakers unraveled the complexities of the Russian codebook

Compiled by T.A. Frail

September 2018

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To get a hint of the monstrously difficult task undertaken by the Venona code breakers, consider all the elaborate steps that Soviet agents took to encrypt a secret message. Here's how it worked, as explained by John Earl Haynes and Harvey Klehr in their book *Venona: Decoding Soviet Espionage in America*. The message—"Pilot delivered report about rockets"—is hypothetical, but it makes reference to an actual American spy, William Ullmann, an Army officer assigned to the Pentagon, whom the Soviets code-named "Pilot." A U.S.-based Soviet agent might send this message to Moscow alerting superiors to check the diplomatic pouch for a dispatch from Pilot.

1. An agent hands the text to a cipher clerk, who uses a code book to convert the words to four-digit numbers:

Pilot	delivered	report	about	rockets
7934	**2157**	**1139**	**3872**	**2166**

2. The clerk shifts one digit to the first group from the second, two digits to the second group from the third, and so on, yielding:

79342	**15711**	**39387**	**22166**	

3. Now the clerk consults a unique "one-time pad." Each page bears 60 five-digit numbers and is supposed to be used just once. At the upper-left corner is a number—26473, in this case—which is inserted

before the first group in the series:

26473	79342	15711	39387	22166

That first number will alert the recipient, who has the same one-time pad, which page of the pad to consult.

4. Then the clerk takes the next four five-digit groups from the one-time pad...

	56328	29731	35682	23798

...and adds them to the four groups that make up the message, using non-carrying arithmetic. (For instance, 8 + 6 = 4, not 14, because nothing is carried):

		79342	15711	39387	22166
+		56328	29731	35682	23798
		25660	34442	64969	45854

Now the entire message looks like this:

26473	25660	34442	64969	45854

5. Next the clerk converts the numerical groups to letter groups, using the formula:

zero=O	1=I	2=U	3=Z	4=T	5=R	6=E	7=W	8=A	9=P

So the message now reads:

UETWZ	UREEO	ZTTTU	ETPEP	TRART

6. The clerk appends another five-letter group (corresponding to the next number from the one-time pad) to signal the end of the message. Finally, he adds a five-digit number, which gives the message a serial number and indicates the date on which it was enciphered. He sends this series of six five-letter words and one five-digit number to Moscow...

7. . . .where another clerk deciphers it, reversing these steps.

Now imagine the Venona team trying to break the code without the benefit of captured Soviet code books or one-time pads.

Source: Venona: Decoding Soviet Espionage in America *by John Earl Haynes and Harvey Klehr (Yale University Press)*